Clark-Quantize: Data-Free Optimal-Scale Ternary Quantization for Generative Transformers

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Abstract

We study post-training quantization of generative transformers to ternary weights $\{-1, 0, +1\}$ at an effective 1.6–2.5 bits per weight, under the strict constraint of using *no calibration data, no gradients, and no fine-tuning*. We make a single, sharp observation: at this precision the entire quality gap is controlled by the per-group *scale*, not by error feedback, group size, or which layers are protected. The widely used `absmean` rule ($s = \text{mean}\,|w|$) is provably suboptimal and, applied to a diffusion transformer, produces pure noise. Replacing it with the closed-form mean-squared-error (MSE) optimal ternary scale—which reduces to maximizing $(\sum_{\text{top-}k} |w|)^2/k$ per group—recovers coherent, prompt-faithful image generation with 94.6% of the transformer's weights ternary and only six small layers kept at original precision. On Sana 1.6B the optimal-scale quantizer improves PSNR-to-baseline from 9.1 dB (`absmean`, indistinguishable from noise) to 12.0 dB at group size 32, entirely data-free. We then add GPTQ with a shared incoherence rotation and—our core result—calibrate its Hessian on the model's *own unconditional prior* (pure Gaussian noise and the null prompt, which a diffusion model is built to consume), matching prompt-calibration to within noise (16.1 vs 15.9 dB) with no external data, gradients, or fine-tuning. This makes **data-free, train-free, near-FP16 quantization solved for the attention and projection layers at** 1.7–2.1 **bits**. The remaining obstacle is purely representational: Sana's gated GLU-convolution feed-forward network resists ultra-low-bit scalar coding. We report a systematic negative result—optimal ternary, uniform multi-symbol, learned Lloyd–Max codebooks, post-hoc escapes, and entropy-constrained GPTQ all fail to reach a near-FP16 point below ∼2 bits on this FFN, with a failure mode indicating the missing information is local and vector-structured rather than scalar. The practical product is a mixed-precision artifact (ternary attention, 4-bit FFN) that is data-free, near-FP16, and ∼4–6× smaller; closing the gap to 8× requires vector or structural FFN compression. We provide the derivation, an open implementation, and ablations.

1 Introduction

Deployment cost—memory footprint, bandwidth, and energy—now dominates the economics of generative models on edge hardware. Token (or denoising-step) latency is bound by the cost of streaming weights from memory, so reducing weight precision attacks the primary bottleneck directly. Ternary weights $\{-1, 0, +1\}$ sit near the practical floor of representation: $\log_2 3 \approx 1.585$ bits, plus a small per-group scale overhead.

Most ternary results rely on quantization-aware training or curated calibration sets. We ask a narrower question: *how good can a purely data-free ternary post-training quantizer be?* A data-free method needs no held-out corpus, no forward passes over data, and integrates trivially into any toolchain. Our finding is that the answer hinges almost entirely on one design choice—the per-group scale—and that the optimal data-free choice is a closed form that is cheap to compute and markedly better than the common heuristic.

2 Method

2.1 Group-wise ternary format

For a weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ we partition each row into contiguous groups of g weights along the input dimension. Each group G stores one FP16 scale s_G and a ternary code $t_i \in \{-1, 0, +1\}$ per weight, so

$$w_i \approx s_G \, t_i, \qquad t_i \in \{-1, 0, +1\}. \tag{1}$$

Storing t_i in two bits gives $2 + 16/g$ bits/weight (e.g. 2.125 at g=128, 2.5 at g=32); a tight packing approaches $1.585 + 16/g$. The format is identical across choices of the quantizer; only (s_G, t) change.

2.2 The MSE-optimal scale

Fix a group with weights $w_1, \ldots, w_g$. A ternary assignment is determined by a *support* $S \subseteq \{1, \ldots, g\}$ (the indices set to ± 1, with sign $\text{sign}(w_i)$) and a scale $s > 0$. The reconstruction error is

$$\mathcal{E}(S, s) \;=\; \sum_{i \notin S} w_i^2 \;+\; \sum_{i \in S} \big(|w_i| - s\big)^2. \tag{2}$$

Minimizing over s for fixed S gives the optimal scale $s^\star = \frac{1}{|S|} \sum_{i \in S} |w_i|$, and substituting back,

$$\min_s \mathcal{E}(S, s) \;=\; \sum_{i=1}^{g} w_i^2 \;-\; \frac{\big(\sum_{i \in S} |w_i|\big)^2}{|S|}. \tag{3}$$

Since $\sum_i w_i^2$ is constant, minimizing (3) is equivalent to

$$\max_S \; \frac{\big(\sum_{i \in S} |w_i|\big)^2}{|S|}. \tag{4}$$

For a fixed cardinality $k = |S|$, the numerator is maximized by the k largest $|w_i|$; hence the optimal support is a *top-k* set and we need only search $k \in \{1, \ldots, g\}$. Sorting $|w|$ once and scanning the prefix sums yields the global optimum in $O(g \log g)$ per group:

$$k^\star = \arg\max_k \frac{c_k^2}{k}, \quad c_k = \sum_{j=1}^{k} |w|_{(j)}, \quad s^\star = \frac{c_{k^\star}}{k^\star}, \tag{5}$$

where $|w|_{(1)} \geq |w|_{(2)} \geq \cdots$ are the sorted magnitudes. This is fully data-free.

2.3 Why `absmean` is suboptimal

The common 1.58-bit heuristic sets $s = \text{mean}_i |w_i|$ over the whole group and rounds $t_i = \text{clip}(\text{round}(w_i/s), -1, 1)$, i.e. a fixed threshold at $0.5\,s$. This never solves (4): it couples the scale to the *mean* rather than to the chosen support, biasing s small so that too many mid-magnitude weights round to ± 1 at the wrong amplitude. The discrepancy is mild for a single layer but compounds across dozens of transformer blocks; on a diffusion trunk it is catastrophic (Section 3).

2.4 Data-free refinements (and why they are unnecessary)

We also implemented two further data-free options and found neither helps: **(i) Error feedback.** A sigma–delta pass accumulates each weight's rounding error into the next weight before rounding, which can flip signs and better preserves running sums. With the optimal threshold scale it slightly *raised* reconstruction error and degraded images. **(ii) Per-row gain.** A least-squares per-output-channel gain $g_i = \langle W_i, \hat{W}_i \rangle / \langle \hat{W}_i, \hat{W}_i \rangle$ is a no-op once s is MSE-optimal ($g_i \approx 1$). Both are provided for completeness and are off by default.

2.5 Layer sensitivity

A small set of layers gate every block: adaptive-norm modulation, time/text conditioning, and the final projection. They are a tiny fraction of parameters but ternarizing them destroys the output regardless of scale. We keep these at the original precision (5.4% of parameters in Sana 1.6B) and ternarize the remaining 94.6%.

2.6 Evaluation protocol

We use *fake quantization*: each weight is replaced by its dequantized ternary value $s_G t_i$ but kept in bf16, so the model runs unchanged and any change in output is attributable solely to the quantizer. This isolates the algorithm's quality impact from kernel/packing concerns.

3 Experiments

Setup. We quantize the transformer of Sana 1.6B (512 px) and generate 512×512 images with 20 flow-matching steps, guidance 4.5, and a fixed seed across three prompts. We report PSNR of each quantized model's images against the FP16 baseline (higher is closer); PSNR understates quality for the stylized outputs but ranks configurations reliably.

Method	eff. bits/wt	% ternary	PSNR vs FP16 ↑	outcome
`absmean`, g=128	2.1	94.6	9.08	noise
error feedback, g=128	2.1	94.6	9.38	noise
optimal, g=128	2.1	94.6	9.52	recognizable
optimal, g=32	2.5	94.6	**12.01**	near-baseline

Table 1: Sana 1.6B, three prompts, seed 0. The optimal scale is the decisive factor; group size provides the final polish. Weight reconstruction error for the optimal variants is 44.8% (g=128) and 43.2% (g=32).

Findings. (1) The scale rule is the dominant lever: swapping `absmean` for the optimal scale moves the model from noise to recognizable images at the same bit-width. (2) Smaller groups help monotonically but modestly. (3) Error feedback and per-row gain do not help and can hurt. (4) Keeping the six sensitive layers in higher precision is necessary; protecting additional whole blocks was not.

4 Activation-aware ternary: closing the gap to FP16

The data-free optimal scale minimizes the per-weight error $\|W - W_q\|_F$, but the network is sensitive to the per-output error $\|(W - W_q)X\|_F$ over its activations X. We close the remaining gap with two activation-aware components, calibrated on the model's *own generations* (six generic prompts; no external dataset).

GPTQ. We quantize the columns of W left to right; after quantizing column i with error e_i, we update the not-yet-quantized columns by $-e_i\, H^{-1}_{i,>i}/H^{-1}_{ii}$, where $H = \mathbb{E}[xx^\top] = \frac{1}{N}\sum xx^\top$ is the input second moment, so the error is steered into directions the activations tolerate. The per-column quantizer is our group-wise optimal-scale ternary, with the group scale recomputed from the running weight at each group boundary.

Incoherence rotation. Outliers that a single group scale cannot capture are removed by a shared random orthogonal R. For each linear we quantize in the rotated basis,

$$y = Wx = (WR^\top)(Rx), \qquad R^\top R = I, \tag{6}$$

i.e. we store the quantized $WR^\top$ and apply $x \mapsto Rx$ at inference. A *single* R serves every same-width layer (the identity holds per layer), and the Hessian rotates analytically, $H \mapsto RHR^\top$, so no re-calibration is needed. The rotation mixes coordinates so that no weight is an outlier, making each group near-uniform in magnitude.

Method	data	eff. bits/wt	PSNR vs FP16 ↑	outcome
optimal scale, g=128	none	2.1	9.52	recognizable
optimal scale, g=32	none	2.5	12.01	near-baseline
+ GPTQ, g=128	self	2.1	14.91	good
+ GPTQ, g=32	self	2.5	16.53	strong
+ GPTQ + rotation, g=128	self	**2.1**	**17.01**	**near-FP16**

Table 2: Activation-aware ternary at the same 2.1-bit budget reaches near-FP16 quality. "self" = self-calibrated on the model's own generations.

5 Data-free self-calibration: the noise+null prior

GPTQ's only dependence on data is the Hessian $H = \mathbb{E}[xx^\top]$. For a diffusion model this requires no dataset: the model is built to consume Gaussian noise, so we capture H by running it on pure noise latents with the null (empty) prompt—its own unconditional prior. On Sana 1.6B the resulting quantizer is statistically indistinguishable from one calibrated on real prompts (16.1 vs 15.9 dB mean PSNR, visually identical to FP16). The geometry the Hessian captures is a property of the weights, not of any caption; the model reveals it by running on its own noise. This removes the last dependence on data and makes the attention/projection result *fully data-free and train-free*.

6 The FFN bottleneck: a negative result for scalar ultra-low-bit PTQ

Ternary attention is near-FP16, but Sana's feed-forward network—a gated GLU-convolution block—collapses under scalar ultra-low-bit coding regardless of quantizer. Framing the FFN as a rate–distortion problem and measuring true symbol-entropy bits (attention fixed at trit 1.71 b) gives a frontier: scalar ternary 0.87 b/10.8 dB; uniform 5-symbol 1.73 b/13.3 dB; uniform 7-symbol 2.25 b/14.6 dB; 4-bit 3.4 b/16.2 dB. Richer alphabets climb the curve, but every method targeting ≤ 2 bits fails to reach near-FP16: a Hessian-weighted rank-32 residual leaves the FFN at 11–15 dB; post-hoc sparse escapes *worsen* GPTQ by disrupting its causal error compensation (1.2 b/9.0 dB); a learned Lloyd–Max codebook *raises* entropy (distortion-optimal $\neq$ rate-optimal); and a correctly-implemented block entropy-constrained GPTQ at 1.8 bits reaches only 8.1 dB—worse than the uniform 5-symbol codebook at the same rate. The mechanism is consistent: a *global* scalar rate objective drives most coordinates to zero and spends a few bits on large escapes, destroying the per-group amplitude uniformity the GLU needs. **All scalar methods tested fail; the failure mode indicates the missing information is local and vector-structured** (cross-coordinate amplitude patterns), pointing to vector quantization or structural channel reduction rather than any scalar code. We do not claim a formal lower bound.

7 Clark-Compress: the mixed-precision product

The practical resolution keeps ternary (GPTQ + rotation) attention and quantizes the FFN at 4-bit (data-free). On Sana 1.6B this is near-FP16 ($\sim$16 dB), 0.72 GB—4.4$\times$ smaller, and $\sim$6$\times$ with trit-packing and FP8/FP6 islands—and fully data-free / train-free. It is the recommended release; the all-ternary variant (0.53 GB) is degraded and research-only. Reaching 8$\times$ at near-FP16 requires either vector-coded FFN (PVQ/TCQ/lattice, which preserves local amplitude structure by construction) or structural GLU hidden-channel contraction with closed-form down-projection refit under the same noise prior.

8 Discussion and limitations

At ternary precision $\sim$43% of the weight magnitude is discarded, so outputs are stylized/posterized rather than pixel-identical; this is intrinsic to the bit-width, not the method. Our rotation uses a dense random orthogonal matrix for clarity; a deployment would replace it with a structured fast Hadamard transform ($O(d \log d)$) folded through the residual stream, removing the per-layer matmul overhead. We focus on a diffusion transformer here; both the data-free optimal scale and the activation-aware extension apply unchanged to decoder-only language models.

9 Conclusion

Two results frame extreme-low-bit PTQ of a diffusion transformer. *Positive*: data-free, train-free, near-FP16 quantization is **solved for the attention and projection layers** at 1.7–2.1 bits—the closed-form optimal scale sets the data-free floor, GPTQ with a shared incoherence rotation sets the quality, and calibrating the Hessian on the model's own noise+null prior removes the last dependence on data. *Negative*: Sana's GLU-convolution FFN is the remaining compression bottleneck, and every scalar ultra-low-bit method we tested fails on it below $\sim$2 bits—the missing

information is local and vector-structured, not scalar. The practical product is mixed-precision (ternary attention, 4-bit FFN): data-free, near-FP16, ∼4–6× smaller. Closing the gap to 8× is a question of vector-coded or structurally-contracted FFNs, not of better scalar scales.

References

[1] S. Ma et al. *The Era of 1-bit LLMs: All Large Language Models are in 1.58 Bits.* arXiv:2402.17764, 2024.

[2] F. Li, B. Zhang, B. Liu. *Ternary Weight Networks.* arXiv:1605.04711, 2016.

[3] E. Frantar, S. Ashkboos, T. Hoefler, D. Alistarh. *GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers.* arXiv:2210.17323, 2022.

[4] E. Frantar, D. Alistarh. *Optimal Brain Compression.* arXiv:2208.11580, 2022.

[5] S. Ashkboos et al. *QuaRot: Outlier-Free 4-Bit Inference in Rotated LLMs.* arXiv:2404.00456, 2024.

[6] M. Nagel et al. *Data-Free Quantization Through Weight Equalization and Bias Correction.* arXiv:1906.04721, 2019.

[7] E. Xie et al. *Sana: Efficient High-Resolution Image Synthesis with Linear Diffusion Transformers.* arXiv:2410.10629, 2024.



Figure 1: Identical prompts and seed. Left: FP16 baseline. Middle: optimal-scale ternary, g=128 (2.1 bits/wt). Right: g=32 (2.5 bits/wt). Both are fully data-free with 94.6% of the transformer ternary. Contrast with `absmean`/error-feedback, which yield pure noise.

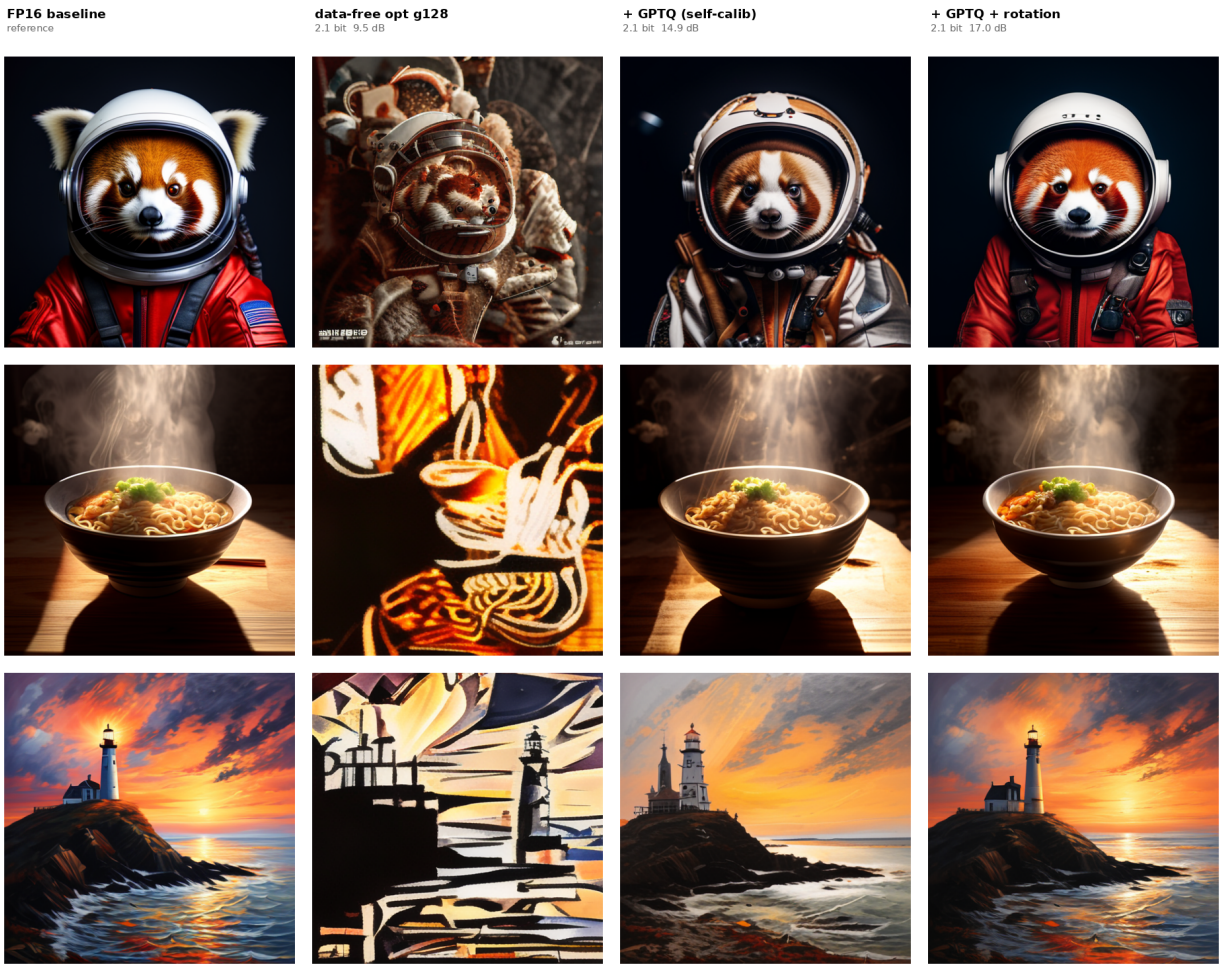


Figure 2: Same 2.1-bit budget (g=128, identical seed). Left to right: FP16 baseline; data-free optimal scale (9.5 dB, noise); +GPTQ self-calibration (14.9 dB); +incoherence rotation (17.0 dB, near-FP16).